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GEOFIT

SMART GEOTHERMAL

WP3 Design Framework for novel ground (slinky/earth basket) type shallow heat exchangers

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D.3.6 Model environment for different types of novel Ground Heat Exchangers

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Executive Summary

The goal of WP3 is to develop an integrated design framework for novel ground (slinky/earth basket) type shallow heat exchangers. This design framework, based on existing and new models of heat transfer and on experimental data, will be implemented in a design- and engineering calculation tool to support the implementation of these new technologies in the market.

This deliverable provides a description of the engineering tool developed in WP3. The purpose of the engineering tool is to provide a consolidated methodology for the design of different ground heat exchangers. The engineering tool provides:

- Calculation of the temperature response of different types of ground heat exchangers to an energy load.
- Methods for sizing of different types of ground heat exchangers.
- Calculation of pressure drop in the ground heat exchanger
- User interface to select and define the ground heat exchanger and associated parameters, present results of the design calculations and a framework for project / design management.

The engineering tool has been developed in the python programming language. The tool has been validated by comparison with other calculation methods and by comparing with data collected in the Geofit project by partner AIT.

The design framework (deliverable D3.2) defined the goals of the (thermal and hydraulic) design (especially sizing) of the ground source heat exchanger, as a function of different boundary conditions (building energy demand, soil thermal parameters, required system performance etc.). The engineering tool provides the calculation methods for the overall system design and will support the engineer in the choices of heat exchanger technology (vertical, horizontal or earth basket/slinky) and other design parameterizations.

This deliverable describes the engineering tool developed in WP3, this version (October 2020) provides background information on the core of the engineering tool, specifically the different calculation modules as well as a first draft of the design project relational database and graphical user interface.

It is the first implementation of a tool that allows all different types of ground heat exchangers to be evaluated in one consolidated environment

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Symbols

$\overline{T}$	:	Mean surface temperature	°C
Ω	:	Angular frequency temperature wave	rad/s
A_o	:	Amplitude temperature wave	K
D	:	Depth below surface	m
d	:	Distance	m
D_h	:	Hydraulic diameter	m
G	:	G-function	-
g	:	Gravitational acceleration	m/s^2
H	:	Borehole length	m
N_{ring}	:	Number of rings	n
P	:	Point	-
q', q_0	:	Heat transfer	W
r	:	Pipe radius, radial distance	m
R	:	Ring radius	m
r_b	:	Borehole radius	m

T	:	Temperature	^oC
t	:	time	s
t_s	:	Characteristic time	s
$T_s(z',t)$	:	Temperature at depth (z') and time (t)	^oC
v	:	Flow rate	m/s
x, y, z	:	Cartesian coordinate	m
z'	:	depth	m
ΔP	:	Pressure drop	Pa
α	:	Thermal diffusivity	m^2s
ε	:	Effective pipe roughness	m
λ	:	Thermal conductivity	W/m·K
ρ	:	Density	kg/m^3
ω, φ	:	Angular parameter	rad

1 Introduction

1.1 background

Different types of ground heat exchangers (GHEX) exist, such as vertical borehole heat exchangers, horizontal ground heat exchangers, slinky type ground heat exchangers and earth-basket or spiral type heat exchangers (D3.1, Design methodologies strengths and weaknesses). All these different type of ground heat exchangers need specialized calculation tools for the design (sizing) and performance evaluation, but so far there are no integrated design tools available to support all these different types of ground heat exchangers. Moreover, no tool exists that supports the overall design process for ground heat exchangers in general.

One of the goals of WP3 is to develop a design framework and engineering tool for novel ground (slinky/earth basket) type shallow heat exchangers. This design framework, based on theoretical models of heat transfer and on experimental data, will be implemented in a design- and engineering calculation tool[1] to support the implementation of these new technologies in the market.

The Geofit deliverables D3.1 (Design methodologies strengths and weaknesses) and D3.2 (Ground Source Heat Exchanger design framework) provide background on the theoretical aspects of ground source heat exchanger calculation methodologies and design process. An integrated fundamental framework for the calculation of the thermal response of different types of ground heat exchangers, all based on the finite line source methodology, has been developed within the framework of WP3 and has been documented (Meeng, 2020: Development of an engineering tool for the design of novel shallow ground heat exchangers – Geofit. Master Thesis TU Eindhoven).

This deliverable provides a description of the engineering tool developed in workpackage 3. The purpose of the engineering tool is to provide a consolidated methodology for the design of different ground heat exchangers. The engineering tool provides:

- Calculation of the temperature response of different types of ground heat exchangers to an energy load.
- Methods for sizing of different types of ground heat exchangers.
- Calculation of pressure drop in the ground heat exchanger
- User interface to select and define the ground heat exchanger and associated parameters, present results of the design calculations and a framework for project / design management.

The main structure and content of the engineering tool is discussed in section 2. Section 3 gives some results of validations that have been performed and section 4 presents the first implementation of the engineering project database and conceptual design of the user interface.

[1] In the context of this report a design tool is more oriented to the "sizing" of a ground source heat exchangers (component sizing) whereas an engineering tool is aimed at the overall system design and will attempt to support the engineer in the choices of heat exchanger technology (vertical, horizontal or earth basket/slinky) and other design parameterisations.

1.2 scope

The engineering tool has been developed in the python programming language python 3.8 (van Rossum, 1995; https://www.python.org/), making it very portable and scalable. Python 3.8 is a modern programming environment with many different modules to aid numerical calculations but also integration with web-servers and databases. It is therefore a very versatile environment for this type of development.

The tool has been validated by comparison with other calculation methods and by comparing with data collected in the Geofit project by partner AIT.

This report documents the engineering tool code and structure of the calculation process and input data needed.

The modules developed (which will be described in more detail in the subsequent paragraphs) are:

- Ghex_design_toolkit_model. This is the main calculation engine with the fundamental analytical solutions (finite line source solutions) for vertical borehole heat exchangers (vBHE), slinky type heat exchangers and earth basket heat exchangers. In this module also the functions to generate G-functions from the fundamental solutions are included.
- Ghex_design_toolkit_topology. Module that calculates the slinky and earth basket topology based on the data provided to the program.
- Ghex_design_toolkit_utility. Module with different utility routines:
    - Functions to read parameter data for different types of GHEX.
    - Function to calculate decompositions and aggregation of energy load profiles.
    - Functions to generate different time-vectors used for load aggregation and G-function generation.
    - Function that calculates the undisturbed vertical ground temperature profile.
    - Function to calculate ground thermal load using temperature-dependent heat pump correlations.
- Ghex_design_toolkit_thermal_resistance. Module to calculate the different thermal resistances needed to obtain fluid temperature from the calculated heat exchanger wall temperature.
- Ghex_design_toolkit_fluid_properties. Module to calculate temperature dependent fluid properties for different mixes of fluid (water, Mono Ethylene Glycol and Mono Propylene Glycol).
- Ghex_design_toolkit_pressure_drop. Module to calculate pressure drops in GHEX pipe systems.

2 Module and function description

2.1 Engineering tool process overview

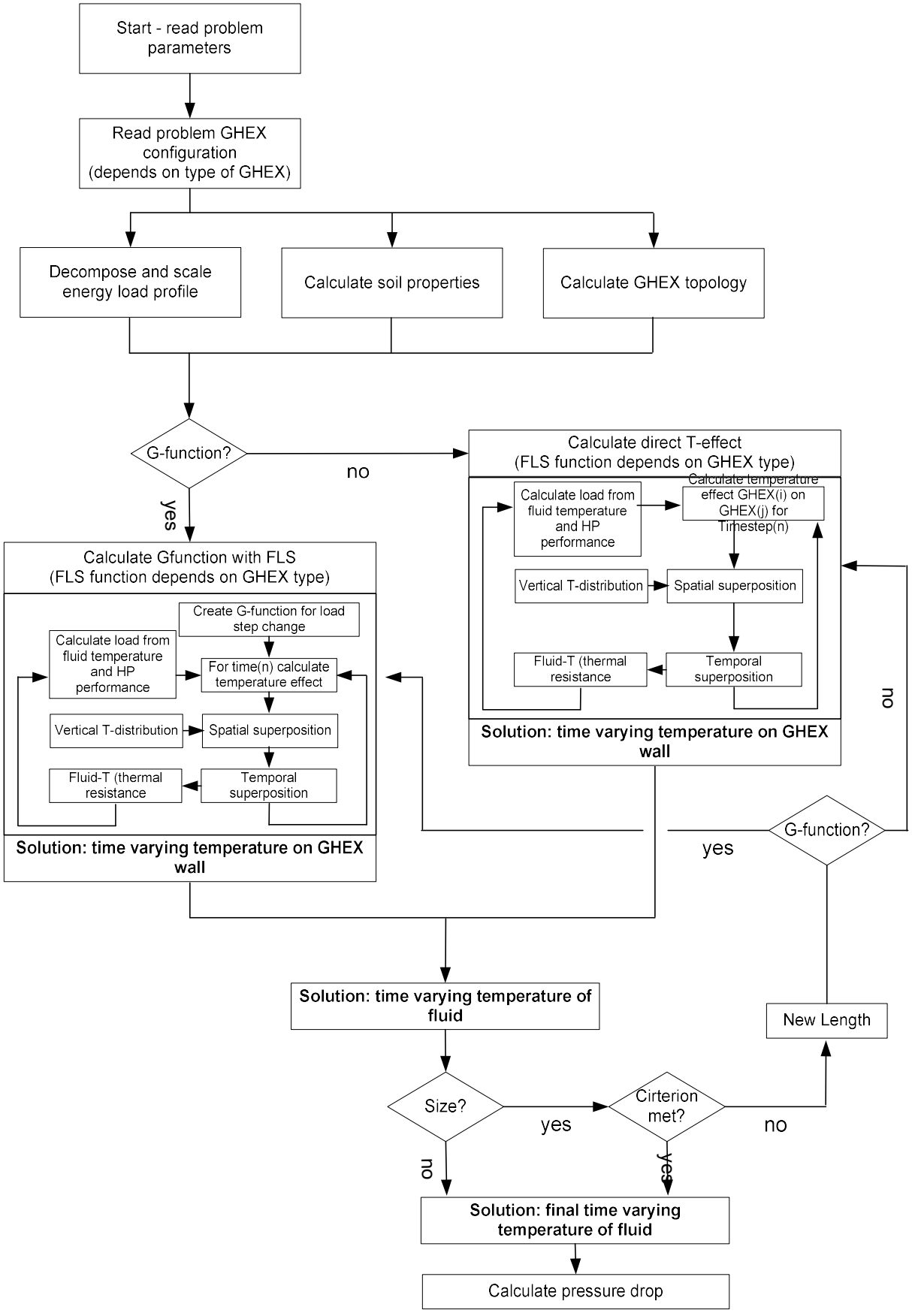
The general flowchart of the engineering tool is depicted in figure 1. First there is a preparation phase where the problem parameters are read from a text file into the program and some preliminary calculations are performed, such as: the energy load profile is decomposed and scaled to energy flux per meter GHEX, soil thermal diffusivity and GHEX topology (distance matrix). Then the main calculation loop is entered. In this loop, for all simulation timesteps the temperature effects are calculated and integrated using spatial and temporal superposition. Here either a direct calculation or G-function approach can be selected. For the direct approach the fundamental finite line source (FLS) model is used to calculate the temperature effects between all individual GHEX components. In the G-function approach, firstly the G-function is generated using a synthetic load-step energy profile after which the G-function is used to calculate the temperature change for all simulation timesteps.

For each time step the loads on the ground heat exchanger are updated based on the calculated fluid temperature (taking into account the thermal resistance between fluid and ground – which may be temperature dependent). With the calculated fluid temperature from the net-load on the ground is recalculated using the temperature dependent heat pump performance curves.

Not shown in this flow chart is the calculation of peak load behaviour, which follows the same basic calculation loop. However, it uses the capacity of the heat pump instead of the energy demand of the system and time steps of hours, not years. The peak load (capacity) temperatures are superposed on the temperatures calculated for the base load (energy).

When all time steps have been processed the final temperature solutions are obtained. If the problem is "sizing" a check is made on the temperature limits set for heating and cooling operation, if the set temperature limits are not achieved a new calculation is started where the size of the GHEX is increased (or decreased). Once the calculations are complete a pressure loss calculation is performed.

Figure 1. Integrated Ground Source Engineering Toolkit (IGSET) general program flowchart.



2.2 Module and function description

The core of the mathematical solutions is based on the MSc thesis of C. Meeng (2020).

2.2.1 Ghex_design_toolkit_model

The methodology for the calculation of the temperature response of GHEX systems is based on the finite line source (FLS) equation (see Meeng 2020 and Geofit-D3.1 for details and background on the methodology). The FLS uses point sources of heat to calculate the temperature after a specified time and at a specified distance from the point locus. By defining series of point sources, with 3D coordinates corresponding to the GHEX topology considered[2], and integrating the solutions of the individual point sources the temperature solution for many complex geometries can be found. The FLS for vertical borehole heat exchangers (1D approach) and slinky type horizontal heat exchangers (2D approach) has been described in literature (see D3.1 and Meeng 2020 and references therein). For earth basket heat exchangers a new 3D FLS method was developed (Meeng 2020). In the engineering tool all three FLS approaches have been implemented in one consolidated algorithmic methodology.

For a ground heat exchanger the earth surface is a boundary where no heat transfer between the GHEX and the surroundings occurs (isothermal boundary). With the FLS method this isothermal boundary at the surface is defined using the method of images, above the surface plane images sources with opposite heat rates are specified and integrated with the FLS solution to define this boundary condition.

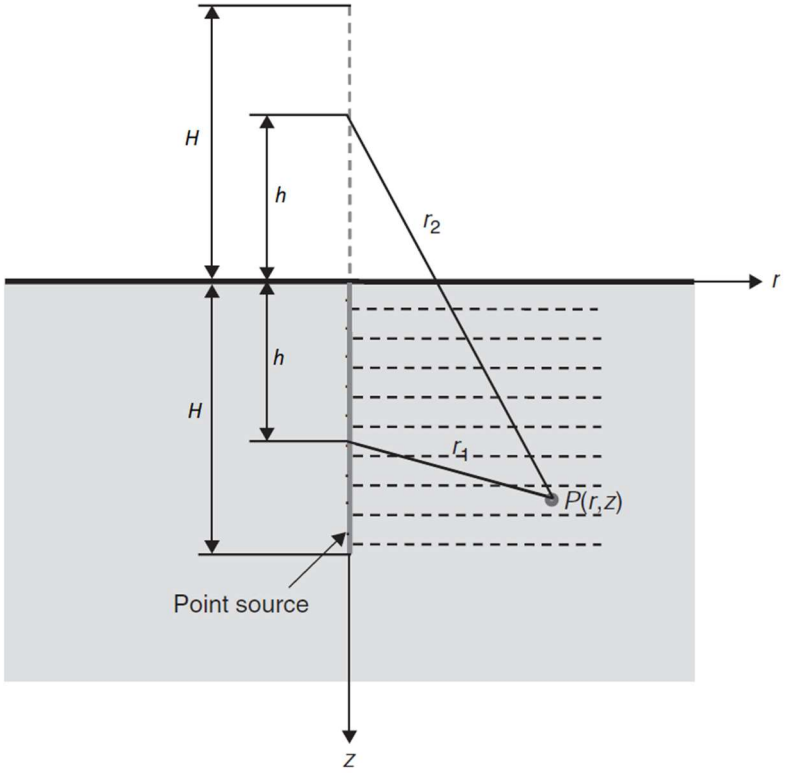
The FLS equations calculate the integral of the temperature effect of a ground heat exchanger on any point (P) in 1, 2 or 3 dimensional space (figures 2 – 5). To solve the design question of either temperature response of the total GHEX or finding the GHEX length required to remain within specified temperature limits, the equations are solved to calculate the temperature effects between the individual heat exchangers in the system.

[2] For a vertical borehole the sequence of 3D points will constitute a line, for an earth basket it will be a spiral shape.

2.2.1.1 Vertical borehole GHEX

The FLS for the vertical borehole heat exchanger and its image source is shown in figure 2.

Figure 2 *Finite Line Source method, line source and image source as used for a Vertical Borehole Heat Exchanger (source: Al-Khoury, 2012).*



The equation that is solved for the vertical borehole heat exchanger FLS is:

$$T(r,z,t) = \frac{q_0}{4\pi\lambda}\int_0^H \left[\frac{1}{r_1} erfc\left(\frac{r_1}{2\sqrt{\alpha t}}\right) - \frac{1}{r_2} erfc\left(\frac{r_2}{2\sqrt{\alpha t}}\right)\right] dh$$

Where the complementary error function *erfc* is: $\frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\,dt$

And the distances r_1 and r_2 to an arbitrary point P(r,z):

$$r_1 = \sqrt{\left(\sqrt{(r^2)} + (z-h)^2\right)} \qquad r_2 = \sqrt{\left(\sqrt{(r^2)} + (z+h)^2\right)}$$

This equation is valid for, with t_s the characteristic time of the field when a steady state condition has been reached:

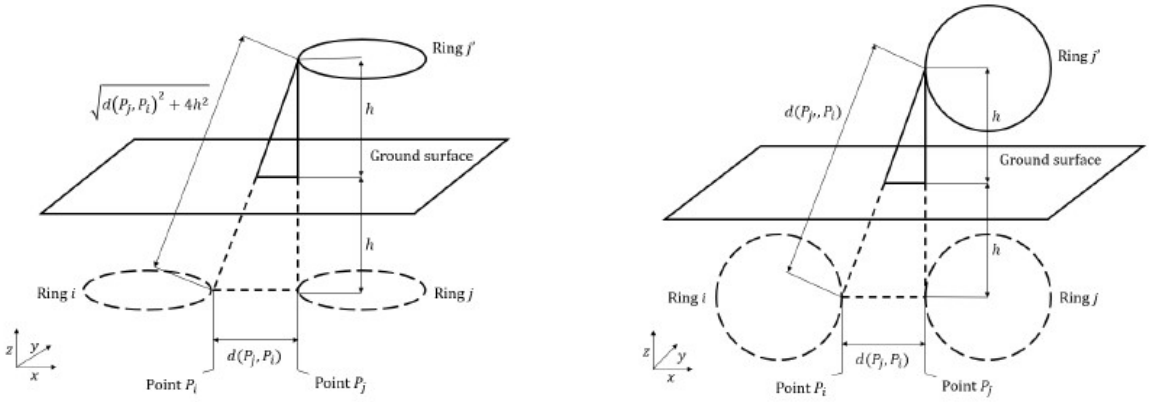
$$5r_b^2/\alpha < t < t_s < 10$$

2.2.1.2 Horizontal and vertical slinky GHEX

The FLS for the horizontal slinky heat exchanger and its image source is shown in figure3a, the vertical in figure 3b.

Figure 3. Finite Line Source method, line source and image source as used for a horizontal (a) and vertical (b) Ground Heat Exchanger (source: Meeng, 2020).

a: horizontal orientation *b: vertical orientation*

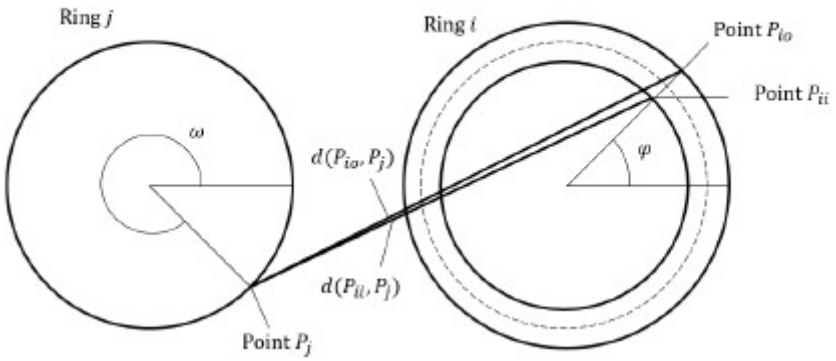


A slinky GHEX consists of one or more rows of one or more rings laid out in a plane that is oriented either vertically or horizontally. Usually the rings in a row overlap. The slinky GHEX is solved by calculating the FLS solution for each ring on all other rings in the GHEX and superposing the solutions in space (and subsequently in time). The equation for the average temperature effect (for the horizontal slinky GHEX) of ring *i* on ring *j* is:

$$\Delta T_{j\rightarrow i}(t) = \frac{q'R}{8\pi^2\lambda}\int_0^{2\pi}\int_0^{2\pi}\left[\frac{\operatorname{erfc}\left(d(P_j,P_i)/2\sqrt{\alpha t}\right)}{d(P_j,P_i)} - \frac{\operatorname{erfc}\left(\sqrt{d(P_j,P_i)^2+4h^2}/2\sqrt{\alpha t}\right)}{\sqrt{d(P_j,P_i)^2+4h^2}}\right]d\omega d\varphi$$

In the equation a double integral is solved to account for the effect of all points of j on all points of i, see figure 4.

Figure 4. Distance between the points on ring i and ring j (source: Meeng, 2020).



The distances are calculated with x_o and y_o as the ring centre coordinates and R the ring radius and r the tube radius:

$$d(P_j, P_i) = \frac{d(P_{ii}, P_j) + d(P_{io}, P_j)}{2}$$

$$d(P_{ii}, P_j) = \sqrt{\left[x_{oi} + (R-r)\cos\varphi - x_{oj} - R\cos\omega\right]^2 + \left[y_{oi} + (R-r)\sin\varphi - y_{oj} - R\sin\omega\right]^2}$$

$$d(P_{io}, P_j) = \sqrt{\left[x_{oi} + (R+r)\cos\varphi - x_{oj} - R\cos\omega\right]^2 + \left[y_{oi} + (R+r)\sin\varphi - y_{oj} - R\sin\omega\right]^2}$$

The equation for the average temperature effect (for the vertical slinky GHEX) of ring i on ring j is in principle the same as for the horizontal heat exchanger, but the expression for the distances is different as the rings are orientated in an x,y,z plane (where for the horizontal slinky the orientation is in the x, y plane):

$$d(P_j, P_i) = \frac{d(P_{ii}, P_j) + d(P_{io}, P_j)}{2}$$

$$d(P_{j'}, P_i) = \frac{d(P_{ii}, P_{j'}) + d(P_{io}, P_{j'})}{2}$$

$$d(P_{ii}, P_j) = \sqrt{\left[x_{oi} + (R-r)\cos\varphi - x_{oj} - R\cos\omega\right]^2 + \left[y_{oi} - y_{oj}\right]^2 + \left[z_{oi} + (R-r)\sin\varphi - z_{oj} - R\sin\omega\right]^2}$$

$$d(P_{io}, P_j) = \sqrt{\left[x_{oi} + (R+r)\cos\varphi - x_{oj} - R\cos\omega\right]^2 + \left[y_{oi} - y_{oj}\right]^2 + \left[z_{oi} + (R+r)\sin\varphi - z_{oj} - R\sin\omega\right]^2}$$

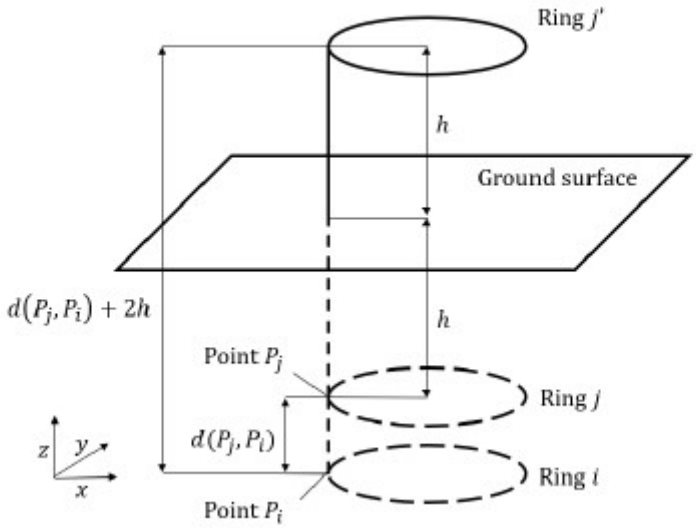
$$d(P_{ii}, P_{j'}) = \sqrt{\left[x_{oi} + (R-r)\cos\varphi - x_{oj} - R\cos\omega\right]^2 + \left[y_{oi} - y_{oj}\right]^2 + \left[z_{oi} + (R-r)\sin\varphi - z_{oj} - 2h - R\sin\omega\right]^2}$$

$$d(P_{io}, P_{j'}) = \sqrt{\left[x_{oi} + (R+r)\cos\varphi - x_{oj} - R\cos\omega\right]^2 + \left[y_{oi} - y_{oj}\right]^2 + \left[z_{oi} + (R+r)\sin\varphi - z_{oj} - 2h - R\sin\omega\right]^2}$$

2.2.1.3 Earth basket GHEX

The earth basket heat exchanger (figure 5) is the most complex type. The earth basket is represented by a stacked sequence of rings at a distance of about 0.1 – 0.5 meters, but there can be many earth baskets in a GHEX field all placed at a distance of between roughly 3 and 10 meters.

Figure 5. Finite Line Source method, line source and image source as used for an earth basket heat exchanger (source: Meeng, 2020).



For the earth basket GHEX the fundamental FLS is the same as for the slinky ring, but the distance calculation is again different:

$$\Delta T_{j\to i}(t)=\frac{q'R}{8\pi^2\lambda}\int_0^{2\pi}\int_0^{2\pi}\left[\frac{\operatorname{erfc}\left(d(P_j,P_i)/2\sqrt{\alpha t}\right)}{d(P_j,P_i)}-\frac{\operatorname{erfc}\left(\left(d(P_j,P_i)+2h\right)/2\sqrt{\alpha t}\right)}{\left(d(P_j,P_i)+2h\right)}\right]d\omega d\varphi$$

$$d(P_j,P_i)=\frac{d(P_{ii},P_j)+d(P_{io},P_j)}{2}$$

$$d(P_{ii},P_j)=\sqrt{\left[x_{oi}+(R-r)\cos\varphi-x_{oj}-R\cos\omega\right]^2+\left[y_{oi}+(R-r)\sin\varphi-y_{oj}-R\sin\omega\right]^2+\left[z_{oi}-z_{oj}\right]^2}$$

$$d(P_{io},P_j)=\sqrt{\left[x_{oi}+(R+r)\cos\varphi-x_{oj}-R\cos\omega\right]^2+\left[y_{oi}+(R+r)\sin\varphi-y_{oj}-R\sin\omega\right]^2+\left[z_{oi}-z_{oj}\right]^2}$$

2.2.1.4 G-function generator

The G-function for the vertical borehole GHEX is generated by:

$$G(t/t_s,r/H,D/H)=\frac{1}{2}\int_{\sqrt{9/(4t/t_s)}}^{\infty}e^{-(r/L)^2s^2}\cdot\frac{Y(s,D/H\cdot s)}{s^2}ds$$

With $s=1/\sqrt{4\alpha(t-t')}$.For the horizontal slinky the G-function equation solved is:

$$G_s(t)=\sum_{i=1}^{N_{ring}}\sum_{j=1}^{N_{ring}}\frac{R}{4\pi N_{ring}}\int_0^{2\pi}\int_0^{2\pi}\left[\frac{\operatorname{erfc}\left(d(P_j,P_i)/2\sqrt{\alpha t}\right)}{d(P_j,P_i)}-\frac{\operatorname{erfc}\left(\sqrt{d(P_j,P_i)^2+4h^2}/2\sqrt{\alpha t}\right)}{\sqrt{d(P_j,P_i)^2+4h^2}}\right]d\omega d\varphi$$

For the vertical slinky the G-function equation solved is:

$$G_s(t) = \sum_{i=1}^{N_{ring}} \sum_{j=1}^{N_{ring}} \frac{R}{4\pi N_{ring}} \int_0^{2\pi} \int_0^{2\pi} \left[\frac{\mathrm{erfc}\left(d\left(P_j, P_i\right)/2\sqrt{\alpha t}\right)}{d\left(P_j, P_i\right)} - \frac{\mathrm{erfc}\left(d\left(\mathrm{P_j}', P_i\right)/2\sqrt{\alpha t}\right)}{d\left(\mathrm{P_j}', P_i\right)} \right] d\omega d\varphi$$

And for the earth basket:

$$G_{eb}(t) = \sum_{i=1}^{N_{ring}} \sum_{j=1}^{N_{ring}} \frac{R}{4\pi N_{ring}} \int_0^{2\pi} \int_0^{2\pi} \left[\frac{\mathrm{erfc}\left(d\left(P_j, P_i\right)/2\sqrt{\alpha t}\right)}{d\left(P_j, P_i\right)} - \frac{\mathrm{erfc}\left(\left(d\left(P_j, P_i\right) + 2h\right)/2\sqrt{\alpha t}\right)}{\left(d\left(P_j, P_i\right) + 2h\right)} \right] d\omega d\varphi$$

2.2.1.5 Computational efficiency and accuracy

The computational speed increases with increasing complexity of the topology and increasing number of individual GHEX in the system. To speed up the calculations two methods have been used:

1) For the direct solution for every time step the solutions are calculated for unique distances only. For distances that have already been calculated a lookup function (hash) is used. In this approach a "unique distance" can be defined with a user-specified threshold so that the calculation for distances that are relatively close together will be treated as one distance.
2) To calculate quickly the effect of many different energy demands G-functions are used. The G-function calculates the typical response to a step change, which is then used to very rapidly calculate the temperature response for the complete energy demand profile.

Table 1 gives an indication of the computational efficiency of the different solution methods. Note that the computational speeds of the direct solutions are calculations of the temperature response for only one year and the generated g-functions can be used to rapidly calculate the temperature response for the lifetime of the GHEX system (30-50 years). A small loss in accuracy is however found while using the G-function method.

Table 1. *Comparison of computational speed for the different solutions and between brackets the speedup factor with regard to direct calculation for one year (The speed is given in seconds but will of course depend on the hardware used, hence the relative "speedup" factors).*

	direct	**direct + hash**	**g-function**
Vertical BHE GHEX	30.6	2.2 (14 x)	1.8 (17 x)
Horizontal Slinky GHEX	13766.4	1815.4 (7.5 x)	1311.7 (10 x)
Vertical Slinky GHEX	13744.8	1931.7 (7 x)	1375.4 (10 x)

2.2.2 Ghex_design_toolkit_topology

The topology module generates the internal data-structure for the complex slinky and earth basket topology based on a few characteristic user supplied parameters (ring diameter, ring pitch, total length amongst others).

2.2.3 Ghex_design_toolkit_utility

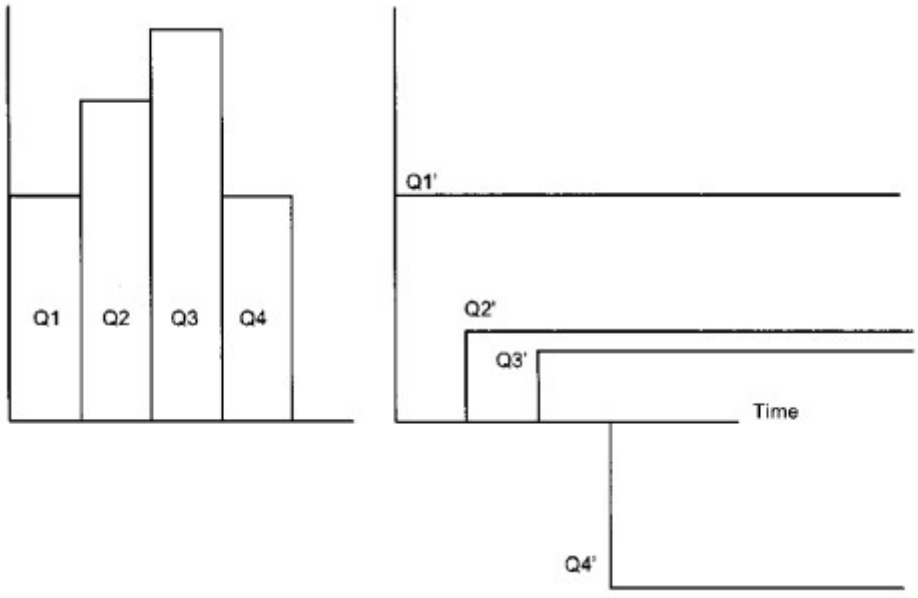
2.2.3.1 Functions to read parameter data for different types of GHEX.

The main simulation data file and GHEX definition files are read in by the program and some checking of parameters is done.

2.2.3.2 Decompositions of energy load profiles.

Both the infinite line source and finite line source solutions calculate the temperature response to a constant heat rate on the GHEX. To be able to calculate time-varying energy demand profiles a decomposition of the energy demand profile in constant heat rates – that add up to the energy demand over longer periods – is used. Usually a monthly energy demand profile is supplied and this is used to calculate the temperature response over periods of 25 – 50 years. The monthly load profile is then repeated for every year. The decomposition principally uses superposition in time of constant heat rates to obtain this response, where the heat rate of the next time step is always the difference between the energy demand of that time step and the aggregated constant heat rates of the previous time steps. As an example consider the case of an energy demand of 10 units in the first month, 20 in the second and 5 in the third month. De decomposed constant heat rates would then be 10, 10 (10 + 10 = 20) and -15 (10 + 10 – 15 = 5). This principle is depicted in figure 6.

Figure 6. Decomposition and superposition in time of an energy demand profile, Q1- Q4 : energy demand for the period, Q1'- Q4': decomposed load profile of constant heat rates that will be superposed in time (Yavuzturk, C. & Spitler, 1999).



2.2.3.3 Functions to generate time-vectors for G-function generation.

To efficiently generate G-functions a geometrically expanding time vector is generated with this function.

2.2.3.4 *Function that calculates the undisturbed vertical ground temperature profile.*

Especially for the slinky and earth basket GHEX the seasonally varying near-surface temperature is important. This temperature is calculated using a well-known analytical function which yields the temperature at a specified depth and time of year. This temperature is added to the solutions calculated with the FLS method.

Input required for the calculation is the average temperature at the surface, the seasonal temperature amplitude, the diffusivity of the soil and the angular frequency of the temperature wave.

$$T_s(z', t) = \bar{T} + A_o e^{-z/D} \sin\left(\Omega t - \frac{z'}{D}\right)$$

Where D:

$$D = \sqrt{\frac{2\alpha_s}{\Omega}}$$

For vertical borehole heat exchangers, extending to greater depth, the geothermal gradient (K/m) can be added.

2.2.3.5 *Function to calculate ground thermal load using temperature-dependent heat pump correlations (to be done).*

When the energy demand is provided as the building energy demand (heating, domestic hot water and cooling), then the energy load on the GHEX is calculated using the efficiency at the operating conditions:

heating energy ground = energy user * ((COP-1) / COP)

cooling energy ground = energy user * (COP/ (COP-1))

where COP (coefficient of performance) is the efficiency (ratio between thermal output and energy input). As the efficiency depends on the GHEX temperature that will vary in time, correlations between the efficiency and temperature (ground heat exchanger) can be supplied.

The efficiency correlations are defined using a second order polynomial fit on temperature (entering water temperature) and efficiency.

Within the calculation the efficiencies are updated when the GHEX temperature changes by a user-specified value (default 5K) or more with regard to the previous calculation.

2.2.4 Ghex_design_toolkit_thermal_resistance.

The equations for calculation of the thermal resistance between fluid and ground for borehole heat exchanger GHEX are given in D3.1 (§ 2.2.2) and are not repeated here.

Some remarks with regard to the implementation:

- For the calculation of the thermal resistance of the fluid in the heat exchanger pipe under laminar conditions the Nusselt correlation of Merker (1987) is used. For turbulent conditions the correlation of Dittus-Boelter (1930) is used.

- For the thermal resistance of the pipe wall a simple formula gives the thermal resistance as a function of wall thermal conductivity and inner/outer diameter.

For horizontal, slinky and earth basket heat exchangers the temperature in the ground is calculated at the pipe wall, the thermal resistance fluid – to – ground then consists of the two correlations above.

For the borehole heat exchanger additional calculations are performed:

- The thermal resistance of the borehole filling is calculated with the best-fit correlation suggested by Liao et al. (2012).
- The internal thermal resistance, accounting for the heat loss between up- and down channel, for a U-loop type heat exchanger is calculated with a first-order multipole approach.
- The effective thermal resistance is calculated following the procedure of Hëllström (1991).

One critical aspect of GHEX thermal resistance is the boundary between laminar and turbulent flow. Usually this is set (arbitrarily) at a Reynolds number of 2300. This results in a very significant change in response of the GHEX system near the boundary: Reynolds number 2299 uses the thermal resistance correlations for laminar flow while a Reynolds number of 2300 uses the correlations for turbulent flow regime (the effect of calculated length of borehole may increase between 15 – 25% with a change of 0.01 l/s in flow rate). To mitigate this effect and achieve a more smooth and realistic representation the thermal resistance is interpolated between a lowest and highest Reynolds number (these can be user-specified). In our calculations we have selected thresholds values for fully laminar regime at a Reynolds < 1000, and for a fully turbulent regime at Reynolds > 3500. In between the thermal resistance is interpolated.

2.2.4.1 Ghex_design_toolkit_fluid_properties.

For the calculation of thermal resistance and efficiency of the heat pump the fluid properties are needed. These are supplied for three different fluid types (water, mono propylene glycol – MPG, and mono ethylene glycol – MEG) for different temperatures and mixing ratios. Although many correlations have been published, in practice using those for the engineering tool would result in an unbalanced and unmanageable algorithm. Therefore we developed a set of second order polynomial fits for the different properties based on data from DOW Chemicals (*https://www.dow.com/content/dcc/en-us/category/market/mkt-building-construction/sub-build-heating-cooling-refrigeration/heat-transfer-fluid-glycol-based-calculator?ffc_type=glycol*).

The freeze point, thermal conductivity, heat capacity, density and viscosity are all calculated using second order polynomial correlations for the pure product which are then expressed for the mixing ratio using mixing rules from Haider-Kahn (2000).

The results were validated with the DOW catalogue for different temperatures and mixing ratios.

2.2.4.2 Ghex_design_toolkit_pressure_drop.

The pressure drop of the GHEX is calculated as a simple pressure drop in a pipe. Other equipment as valves, filters and heat exchangers are not included.

The pressure drop (ΔP) in a pipe is calculated by (Darcy-Weisbach):

$$\Delta P = f\frac{\rho L v^2}{D_h 2}$$

The problem here is to calculate the friction number *f*. For laminar flow the friction number depends only on the Reynolds number (f = 64/Re) but for turbulent flow in rough pipes the Colebrook-White equation is used:

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{\varepsilon}{3.7D_h} + \frac{2.51}{Re\sqrt{f}}\right)$$

Serghide's solution to this equation is very accurate:

$$f = \left[\frac{A-(B-A)^2}{C-(2B)+A}\right]^{-2}$$

With

$$A = -2\log\left[\frac{\left(\varepsilon/D_h\right)}{3.7} + \frac{12}{Re}\right]$$

$$B = -2\log\left[\frac{\left(\varepsilon/D_h\right)}{3.7} + \frac{2.51A}{Re}\right]$$

$$C = -2\log\left[\frac{\left(\varepsilon/D_h\right)}{3.7} + \frac{2.51B}{Re}\right]$$

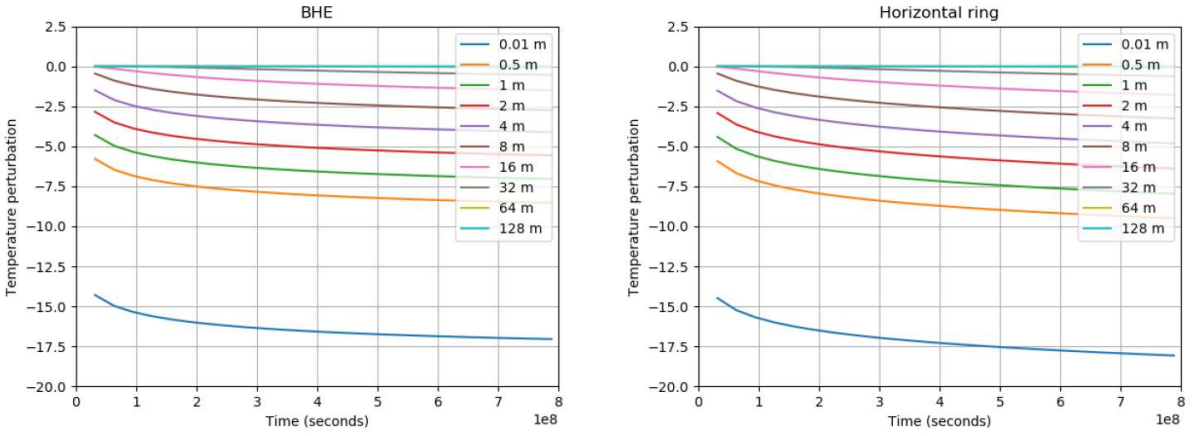
Where ε/D_h is the pipe's relative roughness.

3 Engineering tool code validation

A number of checks and validations were performed during the code development. Many of the checks concern consistency of the code (comparing direct calculation with hash-approach and G-function approach). The code has also been validated against different existing models, mainly engineering calculation tools for vertical borehole heat exchangers and slinky type heat exchangers.

The temperature perturbation due to a single ring source placed in the ground, either horizontally or vertically orientated, is validated against a model that is used for the calculation of temperature perturbation for a BHE with equal length (figure 7). The idea is that the temperature change due to the GHEX between a vertical line source and a circular single ring source should be small for a large ring diameter (similar response). As is clear from the figure, the difference in temperature change is indeed small at larger distances from the source. Similar temperature effects are found using the model for the BHE and the code for the horizontal ring.

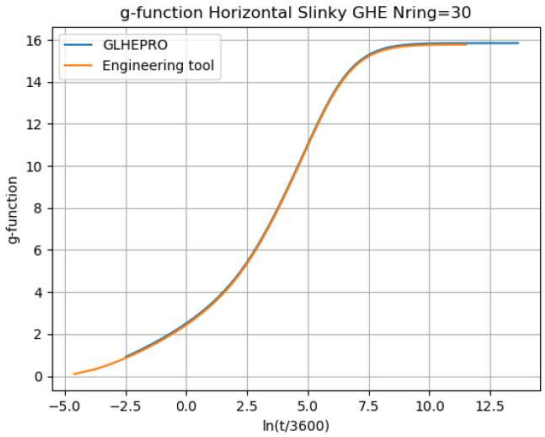
Figure 7. *The temperature perturbation of a BHE compared with the temperature perturbation of a Horizontal ring, length of the BHE 100 meters, diameter of the ring 16 meters (pipe length 100 m).*

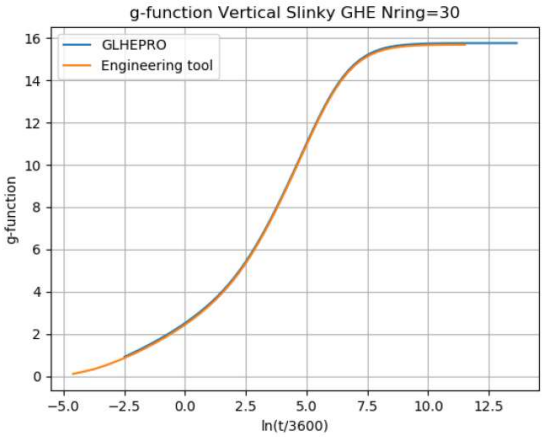


The G-function generated for the slinky type GHEX are validated directly by comparison with another tool named GLHEPRO in which G-functions can be generated for slinky type GHEX. Again very similar results are found (figure 8).

The validation of the earth basket GHEX will done using data from the experiments and CFD simulations performed (AIT, in progress).

Figure 8. G-Functions for Slinky GHEXs generated with the GLHEPRO tool and the engineering tool.





4 Engineering tool project database and user interface

Next step in developing the engineering tool is to provide a user/project database and graphical user interface to the database from which the calculation tool can be called and output can be viewed.

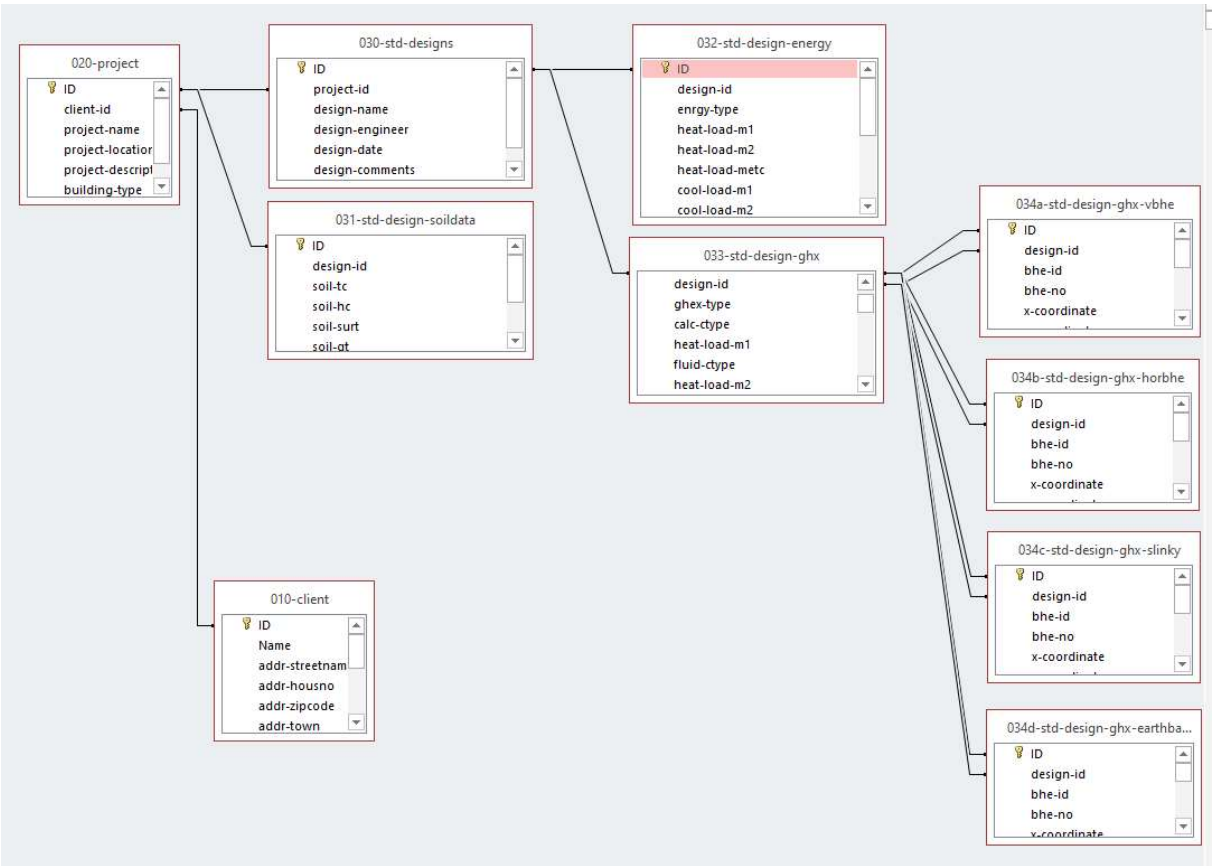
4.1 Database structure

For the management of the design process and the user-interface a first design is being developed. The data will be stored in a relational database (layout shown in figure 9).

The database is centred on project: projects are owned by a user (client). Within a project different designs can be made, where every design can have a specific energy demand profile and/or can have one or more calculations for one or more different GHEX types.

This means that within one project comparisons can be made for one type of GHEX with different energy demand profiles, for one energy profile with different GHEX options or a combination.

Figure 9. General database structure project – design database engineering tool



4.2 User interface definition

The user interface is a web-based interface to view, add, edit and report on the designs made in a project. The user interface will connect to the engineering toolbox calculation engine to perform the actual design calculations.

The Graphical User Interface (GUI) has a simple layout (figure 10 - 12) with a main toolbar along the top and a secondary menu along the left side. Information is displayed and can be edited in the central screen.

A few examples are shown below.

Figure 10. GUI - Welcome screen

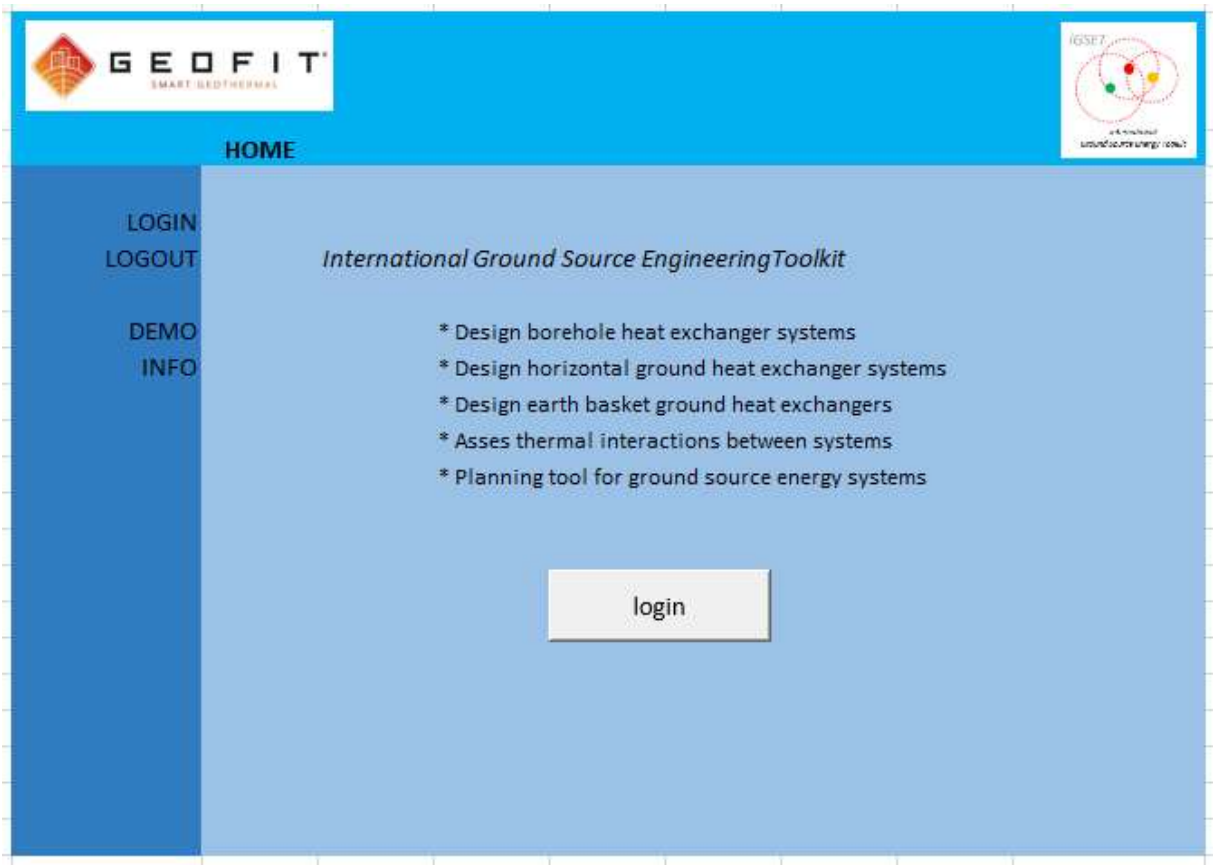


Figure 11. GUI - Overview of client/user projects (after login)

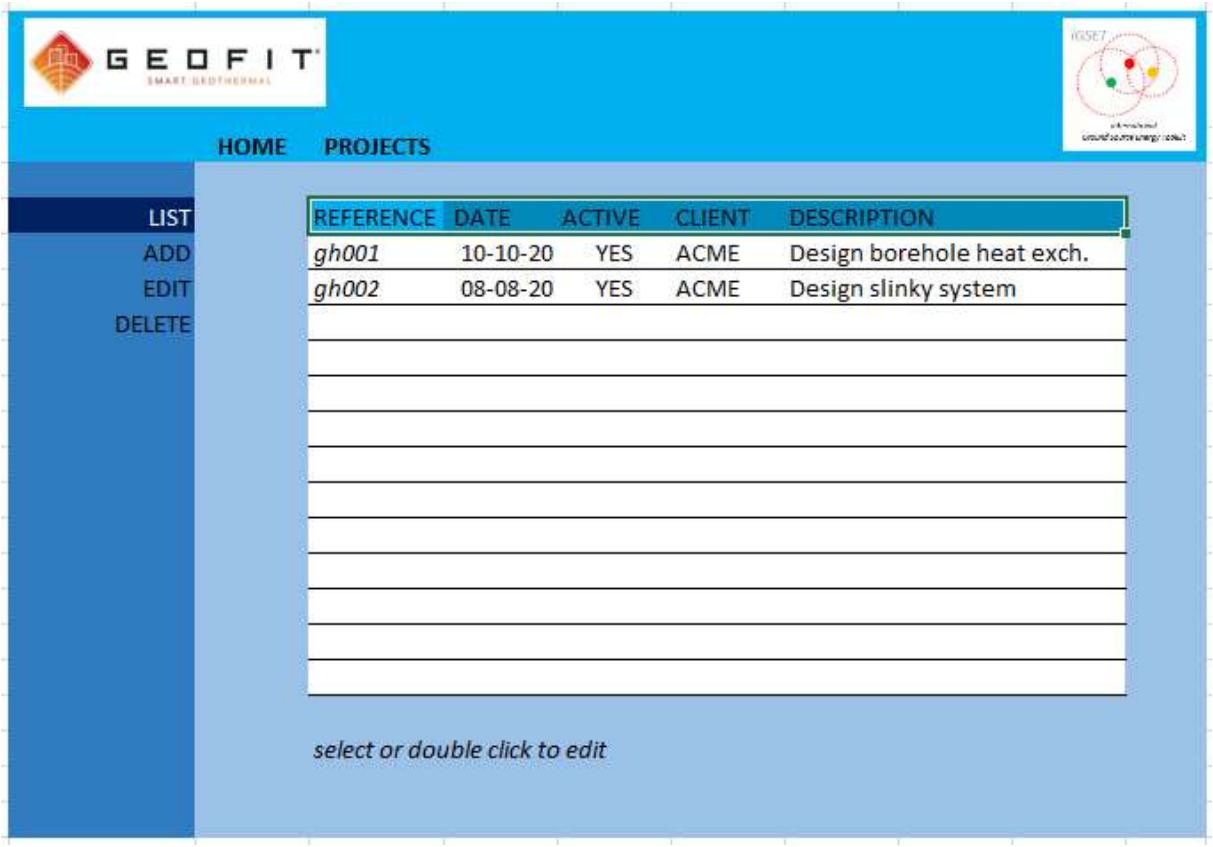


Figure 12. GUI - Selected project showing energy demand profiles and different GHEX designs



5 Conclusion

This deliverable describes the first version of the engineering tool developed within the framework of the Geofit project. The engineering tool allows a designer and engineer to design different types of ground heat exchangers (vertical boreholes, horizontal ground heat exchangers, slinky and earth basket ground heat exchangers). It is the first implementation of a tool that allows all different types of ground heat exchangers to be evaluated in one consolidated environment.

The code has been developed in the programming language python 3.8, making it very portable and scalable. Python 3.8 is a modern programming environment with many different modules to aid numerical calculations but also integration with web-servers and databases. It is therefore a very versatile environment for this type of development.

All different GHEX parameters needed for the calculation are supplied in a text file which is parsed by the program.

The tool has been based (albeit in a completely new implementation) partly on existing state-of-the-art design methodologies, such as:

- Finite line source approach for vertical borehole and slinky type heat exchangers.
- Decomposition of load data and calculation of G-functions for increased computational efficiency.
- Fluid-to-ground thermal resistance calculations.
- Calculation of vertical temperature profile in the ground volume.

New methods have been developed and implemented as well:

- Finite line source approach for earth basket (spiral) heat exchangers.
- Increased computational efficiency by using hashes of distance matrices.
- Implementation of a smooth transition from laminar to turbulent flow regimes.
- Automated temperature dependent fluid properties calculations (based on newly developed fluid property calculations)
- Heat pump efficiency correlations to calculate the GHEX energy loads using the efficiency of the heat pump at different temperatures.
- Pressure drop calculations integrated in the design tool itself.

In the next period especially the user interface and project-database will be further developed and integrated with the engineering tool.

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